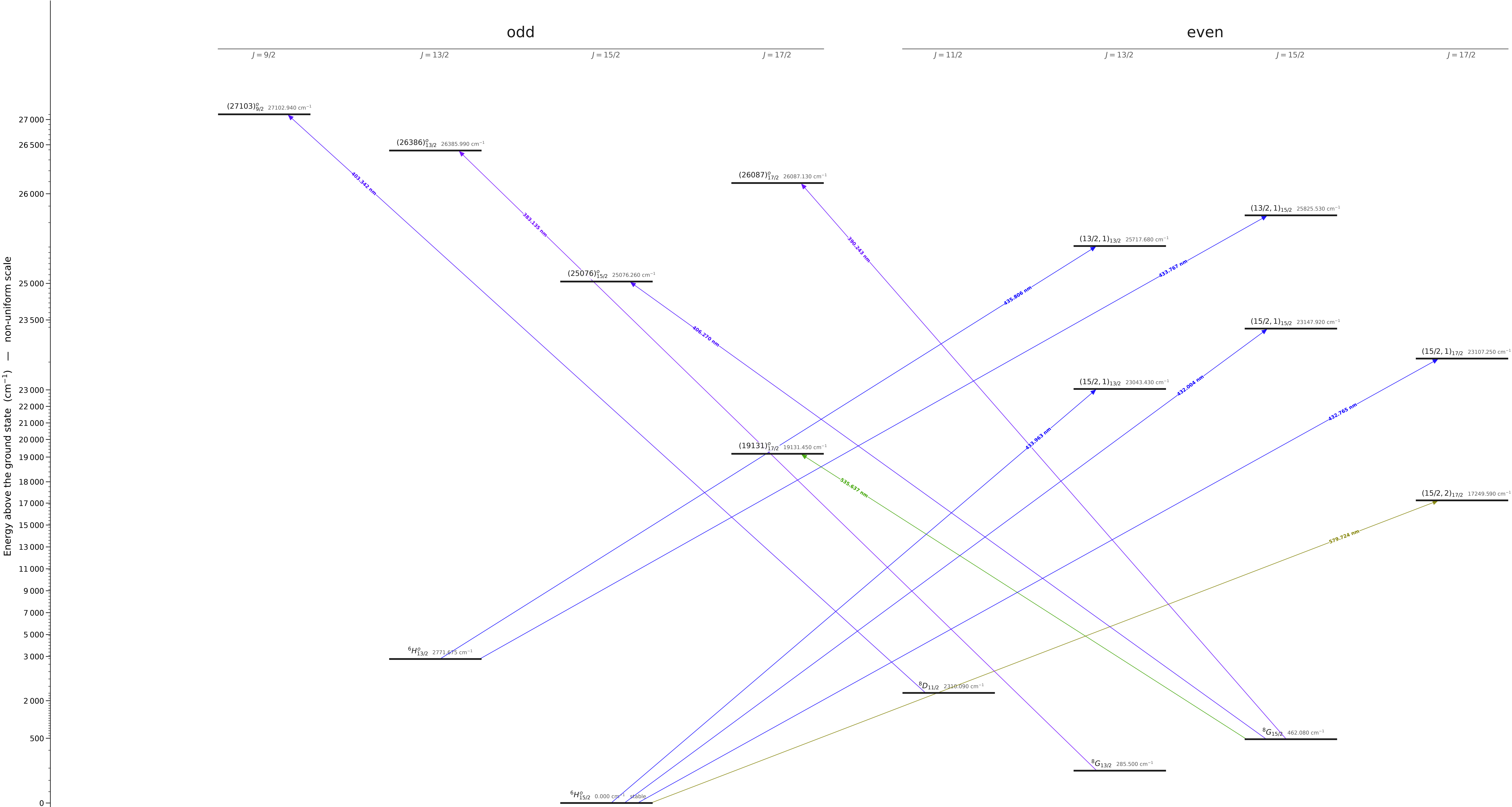
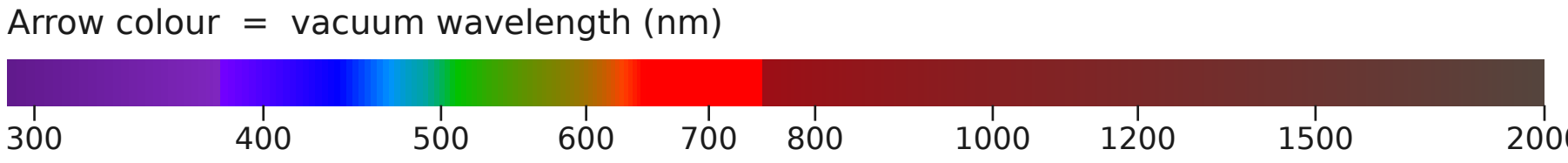


Terbium

Tb (Tb I). Strongest classified transitions below 2 μm (NIST): 16 levels, 11 lines.

Tb⁺ ionisation limit 47295.00 cm⁻¹ = 5.86383 eV (211.439 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$^6H_{13/2}^o - (15/2, 2)_{17/2}$	579.7239	579.5632	517.1297	-	-	-	-	first classified line between 433.8 and 1023.9 nm that co...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁵⁹Tb hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^6H_{15/2}^o$	673.7550(10)	1449.445(12)	F=6: -8062.36 F=7: -4215.75 F=8: +1008.64 F=9: +7942.10
$^8G_{13/2}$	532.2040(20)	928.861(20)	F=5: -5630.04 F=6: -3008.42 F=7: +591.97 F=8: +5421.20
$^8G_{15/2}$	472.6430(20)	1154.239(17)	F=6: -5605.73 F=7: -2989.77 F=8: +659.46 F=9: +5605.79
$^8D_{11/2}$	405.106(3)	-92.64(5)	F=4: -3988.10 F=5: -1903.62 F=6: +542.17 F=7: +3318.97
(15/2, 2) _{17/2}	857.1(4)	1758(5)	F=7: -11599.67 F=8: -5776.98 F=9: +1762.41 F=10: +11367.52
(15/2, 1) _{13/2}	685.7(20)	915(24)	F=5: -7362.20 F=6: -3811.08 F=7: +865.65 F=8: +6914.32

Measured A, B constants; sources in the data file.